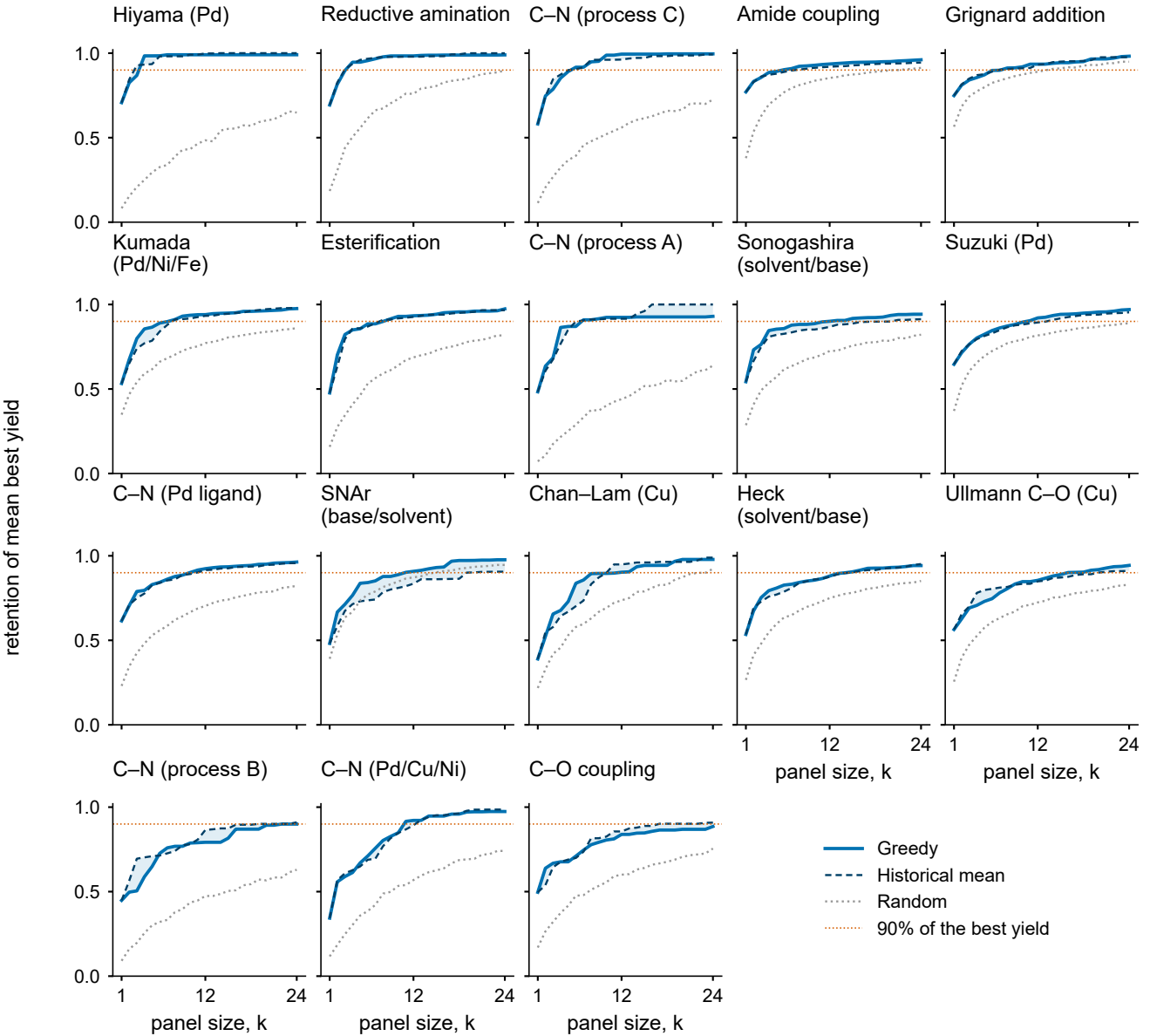
Hiyama (Pd)
Reductive amination
C–N (process C)
Amide coupling
Grignard addition
Kumada (Pd/Ni/Fe)
Esterification
C–N (process A)
Sonogashira (solvent/base)
Suzuki (Pd)
C–N (Pd ligand)
SNAr (base/solvent)
Chan–Lam (Cu)
Heck (solvent/base)
Ullmann C–O (Cu)
C–N (process B)
C–N (Pd/Cu/Ni)
C–O coupling
retention of mean best yield
1.0
0.5
0.0
1
12
24
panel size, k
Greedy
Historical mean
Random
90% of the best yield